Embedded Protection System — Progress Report

Three-phase pump power abnormality monitoring and protection · Personal engineering portfolio

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1. Project overview

The project is a low-voltage embedded monitoring and protection prototype for a three-phase submersible pump application. It is intended to measure current and voltage on all three phases, identify configured electrical abnormalities, and command a relay output intended for a contactor-coil interface. A manual reset is required after a trip. The project is documented as portfolio work relevant to embedded systems.

2. Hardware platform

Category	Selected component / configuration	Role / notes
Controller	ESP32-WROOM-32	Control logic, display, GPIO and internal ADC inputs
External ADC	ADS1115	16-bit ADC used for the three current-sensor channels over I²C
Current sensing	3 × SCT-013, 100 A / 50 mA	Current transformer outputs; burden and bias circuit to ADS1115
Voltage sensing	3 × ZMPT101B modules	Voltage measurement modules connected to ESP32 GPIO 35, 32 and 33
Output	5 V relay module, TONGLING JQC-3FF	Relay command output; opto-isolated module as specified in project notes
User interface	0.96-inch SSD1306 OLED, 3 LEDs, reset button	Status indication and manual reset
Low-voltage power	ATX 5 V supply with 1 A fuse	Fused low-voltage supply; decoupling specified

3. Reported implementation status

Work item	Status reported on 3 October 2026
Low-voltage supply, fuse and decoupling	Completed
ESP32, OLED, LEDs, button and relay assembly	Completed
Three current-sensor channels with burden/bias circuits	Completed
Three voltage-sensor modules	Completed
ADS1115 integration and six-channel routing	Completed
Basic protection logic, startup delay and trip/reset flow	Completed
Calibration-friendly firmware structure	Completed
Real-pump calibration and contactor interface	Next; not reported as completed

4. Sensor assignment

Measured signal	Sensor	Connection
Current — Phase 1	SCT-013, with burden/bias	ADS1115 AIN0
Current — Phase 2	SCT-013, with burden/bias	ADS1115 AIN1
Current — Phase 3	SCT-013, with burden/bias	ADS1115 AIN2
Voltage — Phase 1	ZMPT101B	ESP32 GPIO 35
Voltage — Phase 2	ZMPT101B	ESP32 GPIO 32
Voltage — Phase 3	ZMPT101B	ESP32 GPIO 33

5. Firmware features reported

• Live readings for three current and three voltage channels
• OLED states: OK, STARTING and TRIPPED
• Over-current threshold currently configured at 32 A
• Over/under-voltage window currently configured at 180–250 V
• Four-second startup delay intended to avoid nuisance trips during motor inrush
• Trip state indicated by red LED and relay activation
• Manual reset through push button
• Calibration factors grouped near the top of the code

These are reported prototype features and configured values, not evidence of validated protective performance. Thresholds and measurement accuracy require calibration and testing against suitable references.

6. Remaining work

1. Calibrate current and voltage measurements using appropriate test equipment.
2. Review sensor isolation, module limits, fusing, wiring, grounding and enclosure before any field use.
3. Have the proposed relay/contactor interface reviewed and approved by qualified personnel before connection.
4. Conduct staged tests, then document results and limitations before any live-pump trial.
5. Complete code cleanup, test evidence and repository documentation.

7. Scope and limitations

This report covers the embedded low-voltage monitoring/protection prototype only. It does not document or claim completion of the power-panel assembly, a validated safety function, certification, or successful live-pump testing. The system must not be treated as the sole protective measure for people or equipment.

Revision note: standardized terminology and sensor assignments; removed inconsistent job-title wording and company-branded footer; clarified portfolio status and approval requirements.